HDOC Day-38

String Operations - Part 3

Question: Write a menu-driven C program using separate functions for the operations shown in the assignment (options 1-3 and 4 to exit).

C Program:

```
#include <stdio.h>

void countTypes(char s[]) {
    int i, spaces = 0, digits = 0, special = 0;
    char ch;
    for (i = 0; s[i] != '\0'; i++) {
        ch = s[i];
        if (ch == ' ') spaces++;
        else if (ch >= '0' && ch <= '9') digits++;
        else if (!((ch >= 'a' && ch <= 'z') ||
                   (ch >= 'A' && ch <= 'Z'))) special++;
    }
    printf("Spaces = %d, Digits = %d, Special = %d\n",
           spaces, digits, special);
}

void frequency(char s[]) {
    char ch;
    int i, count = 0;
    printf("Enter character to find: ");
    scanf(" %c", &ch);
    for (i = 0; s[i] != '\0'; i++)
        if (s[i] == ch) count++;
    printf("Frequency of %c = %d\n", ch, count);
}

void toggleCase(char s[]) {
    int i;
    for (i = 0; s[i] != '\0'; i++) {
        if (s[i] >= 'a' && s[i] <= 'z')
            printf("%c", s[i] - 32);
        else if (s[i] >= 'A' && s[i] <= 'Z')
            printf("%c", s[i] + 32);
        else printf("%c", s[i]);
    }
    printf("\n");
}

int main() {
    char s[200];
    int choice, i;
    printf("Enter a string: ");
    fgets(s, sizeof(s), stdin);
    for (i = 0; s[i] != '\0'; i++) {
        if (s[i] == '\n') { s[i] = '\0'; break; }
    }
    do {
        printf("\n1. Count spaces, digits and special characters");
        printf("\n2. Character frequency");
        printf("\n3. Toggle case");
        printf("\n4. Exit\nEnter choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1: countTypes(s); break;
            case 2: frequency(s); break;
```

```
            case 3: toggleCase(s); break;
            case 4: printf("Exit\n"); break;
            default: printf("Invalid choice\n");
        }
    } while (choice != 4);
    return 0;
}
```

Sample Output: For input HeLlo 12!: spaces = 1, digits = 2, special = 1; frequency of l = 1; toggled case = hElLO 12!.